```
#include <iostream>
using namespace std;

int referenceString[100], RS, FR;

void show(int f[], int n)
{
   cout << " Frames: ";
   for (int i = 0; i < n; i++)
   {
      if (f[i] == -1)
         cout << "- ";
      else
         cout << f[i] << " ";
   }
   cout << "\n";
}

void fifo()
{
   int f[100], ptr = 0, faults = 0;

   for (int i = 0; i < FR; i++)
      f[i] = -1;

   cout << "\n===== FIFO =====\n";

   for (int i = 0; i < RS; i++)
   {
      int hit = 0;

      for (int j = 0; j < FR; j++)
         if (f[j] == referenceString[i])
            hit = 1;

      if (!hit)
      {
         f[ptr] = referenceString[i];
         ptr = (ptr + 1) % FR;
         faults++;
         cout << "Ref " << referenceString[i] << " -> FAULT";
      }
      else
      {
```

```
            cout << "Ref " << referenceString[i] << " -> hit ";
        }

        show(f, FR);
    }

    cout << "Total Page Faults (FIFO) = " << faults << "\n";
}

void lru()
{
    int f[100], last[100], faults = 0;

    for (int i = 0; i < FR; i++)
    {
        f[i] = -1;
        last[i] = -1;
    }

    cout << "\n===== LRU =====\n";

    for (int i = 0; i < RS; i++)
    {
        int hit = -1;

        for (int j = 0; j < FR; j++)
            if (f[j] == referenceString[i])
                hit = j;

        if (hit == -1)
        {
            int pos = 0;

            for (int j = 1; j < FR; j++)
                if (last[j] < last[pos])
                    pos = j;

            f[pos] = referenceString[i];
            last[pos] = i;
            faults++;
            cout << "Ref " << referenceString[i] << " -> FAULT";
        }
        else
        {
```

```
            last[hit] = i;
            cout << "Ref " << referenceString[i] << " -> hit ";
        }

        show(f, FR);
    }

    cout << "Total Page Faults (LRU) = " << faults << "\n";
}

void optimal()
{
    int f[100], faults = 0;

    for (int i = 0; i < FR; i++)
        f[i] = -1;

    cout << "\n===== Optimal =====\n";

    for (int i = 0; i < RS; i++)
    {
        int hit = 0;

        for (int j = 0; j < FR; j++)
            if (f[j] == referenceString[i])
                hit = 1;

        if (!hit)
        {
            int pos = -1, farthest = -1;

            for (int j = 0; j < FR; j++)
            {
                if (f[j] == -1)
                {
                    pos = j;
                    break;
                }

                int k;
                for (k = i + 1; k < RS; k++)
                    if (f[j] == referenceString[k])
                        break;
```

```
            if (k == RS)
            {
                pos = j;
                break;
            }

            if (k > farthest)
            {
                farthest = k;
                pos = j;
            }
        }

        f[pos] = referenceString[i];
        faults++;
        cout << "Ref " << referenceString[i] << " -> FAULT";
    }
    else
    {
        cout << "Ref " << referenceString[i] << " -> hit ";
    }

    show(f, FR);
  }

  cout << "Total Page Faults (Optimal) = " << faults << "\n";
}

int main()
{
  cout << "Enter number of frames: ";
  cin >> FR;

  if (FR <= 0 || FR > 100)
  {
    cout << "Invalid number of frames\n";
    return 0;
  }

  cout << "Enter length of reference string: ";
  cin >> RS;

  if (RS <= 0 || RS > 100)
  {
```

```
        cout << "Invalid reference string length\n";
        return 0;
    }

    cout << "Enter reference string: ";
    for (int i = 0; i < RS; i++)
        cin >> referenceString[i];

    fifo();
    lru();
    optimal();

    return 0;
}
```

```
siyasinnarkar@Siyas-MacBook-Pro-2 os % g++ page_replace.cpp -o page_replace
siyasinnarkar@Siyas-MacBook-Pro-2 os % ./page_replace
Enter number of frames: 4
Enter length of reference string: 11
Enter reference string: 7 0 2 3 0 2 1 4 3 2 8

===== FIFO =====
Ref 7 -> FAULT Frames: 7 - - -
Ref 0 -> FAULT Frames: 7 0 - -
Ref 2 -> FAULT Frames: 7 0 2 -
Ref 3 -> FAULT Frames: 7 0 2 3
Ref 0 -> hit  Frames: 7 0 2 3
Ref 2 -> hit  Frames: 7 0 2 3
Ref 1 -> FAULT Frames: 1 0 2 3
Ref 4 -> FAULT Frames: 1 4 2 3
Ref 3 -> hit  Frames: 1 4 2 3
Ref 2 -> hit  Frames: 1 4 2 3
Ref 8 -> FAULT Frames: 1 4 8 3
Total Page Faults (FIFO) = 7

===== LRU =====
Ref 7 -> FAULT Frames: 7 - - -
Ref 0 -> FAULT Frames: 7 0 - -
Ref 2 -> FAULT Frames: 7 0 2 -
Ref 3 -> FAULT Frames: 7 0 2 3
Ref 0 -> hit  Frames: 7 0 2 3
Ref 2 -> hit  Frames: 7 0 2 3
Ref 1 -> FAULT Frames: 1 0 2 3
Ref 4 -> FAULT Frames: 1 0 2 4
Ref 3 -> FAULT Frames: 1 3 2 4
Ref 2 -> hit  Frames: 1 3 2 4
Ref 8 -> FAULT Frames: 8 3 2 4
Total Page Faults (LRU) = 8

===== Optimal =====
Ref 7 -> FAULT Frames: 7 - - -
Ref 0 -> FAULT Frames: 0 - - -
Ref 2 -> FAULT Frames: 0 2 - -
Ref 3 -> FAULT Frames: 0 2 3 -
Ref 0 -> hit  Frames: 0 2 3 -
Ref 2 -> hit  Frames: 0 2 3 -
Ref 1 -> FAULT Frames: 1 2 3 -
Ref 4 -> FAULT Frames: 4 2 3 -
Ref 3 -> hit  Frames: 4 2 3 -
Ref 2 -> hit  Frames: 4 2 3 -
Ref 8 -> FAULT Frames: 8 2 3 -
Total Page Faults (Optimal) = 7
siyasinnarkar@Siyas-MacBook-Pro-2 os %
```